SIMATS ENGINEERING

ASSESSMENT TOOL 1: Agent Design Assessment

COURSE NAME: ARTIFICIAL INTELLIGENCE AND EXPERT SYSTEMS COURSE CODE: MLA01

COURSE OUTCOME COVERED

CO1: *Apply search algorithms and heuristic techniques to solve state-space search and optimization problems.* (BTL 3)

Assessment Tool Weightage: 55 %

Total Marks: 50 **Time: 60 Mins**
No. of Questions: 1

Annexure A

Agent Design Assessment

Code of Conduct:

I _____S.Venkateswarlu__ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: S. Venkateswarlu

Question	Problem Understanding & AI Concept Identification	Dialogflow Agent Design & Implementation	Problem Formulation & Conversation Flow Design	Application of AI Techniques & Reasoning	Testing, Evaluation & Documentation	Total
Weightage	20%	30%	20%	20%	10%	100%
Q1						

Signature of the Faculty Total Marks Awarded

Scenario

A university wants to develop an **AI-based Student Support Assistant** to provide instant responses to student queries related to course registration, timetable, examination schedules, attendance details, and campus services.

As an AI developer, design and implement a conversational agent using **Dialogflow** that can understand user requests, process intents, and provide appropriate responses.

Q. No.	Criteria to be evaluated	BL	Marks
1	**Problem Analysis and Agent Design:** Analyze the student support problem and explain how Artificial Intelligence can solve the problem. Design the conversational agent by defining its objective, users, environment, inputs, outputs, and expected performance measures.	BL3	10
2	**Dialogflow Agent Development:** Create a Dialogflow agent by designing appropriate Intents, Entities, Training Phrases, Responses, and Contexts. Explain how the agent perceives user queries and generates responses. Provide screenshots of the implementation.	BL5	10
3	**Problem Formulation and Decision Flow:** Represent the chatbot interaction as a problem-solving process by defining the initial state, user goal, possible actions, state transitions, and goal completion conditions. Draw the conversation flow diagram.	BL6	10
4	**Search Strategy Application:** Analyze how search techniques used in AI problem solving can support conversational agents. Apply a suitable search approach (such as Breadth First Search or heuristic-based search) to model intent selection or dialogue navigation and justify your choice.	BL5	10
5	**Testing and Evaluation:** Test the Dialogflow agent with different user queries. Evaluate accuracy, response quality, handling of unknown queries, and suggest improvements.	BL6	10

Evaluation Rubrics

Criteria	Weightage (%)	Excellent (Full Marks)	Good	Average	Poor
Problem Understanding & AI Concept Identification	20%	10 – Clearly analyzes the student support problem, defines AI application, agent objective, users, environment, inputs, outputs, and performance measures with proper justification.	7.5 – Identifies most components of the agent design with minor omissions.	5 – Provides basic problem analysis with limited explanation of agent components.	0–2.5 – Incomplete understanding of problem and agent design.
Dialogflow Agent Design & Implementation	30%	10 – Successfully creates a functional Dialogflow agent with appropriate intents, entities, training phrases, responses, contexts, and explains interaction flow with screenshots.	7.5 – Develops major Dialogflow components with minor configuration errors.	5 – Creates basic intents and responses with limited entity/context utilization.	0–2.5 – Incomplete or incorrect Dialogflow implementation.
Problem Formulation & Conversation Flow Design	20%	10 – Clearly represents chatbot interaction using initial state, goal state, actions, state transitions, goal conditions, and a well-designed conversation flow diagram.	7.5 – Provides a structured flow with minor missing details.	5 – Provides basic flow representation with limited explanation.	0–2.5 – Poor or missing problem formulation and decision flow.
Application of AI Techniques & Reasoning	20%	10 – Correctly analyzes and applies suitable search strategies (BFS/heuristic search) for intent selection or dialogue navigation with strong justification.	7.5 – Applies appropriate search strategy with reasonable explanation.	5 – Provides basic understanding of search strategy application.	0–2.5 – Unable to relate search concepts with conversational agent.
Testing, Evaluation & Documentation	10%	10 – Performs comprehensive testing using multiple user queries, evaluates accuracy, response quality, unknown query handling, and suggests meaningful improvements.	7.5 – Performs adequate testing and provides evaluation with minor gaps.	5 – Conducts limited testing with basic observations.	0–2.5 – Insufficient testing and evaluation.

1.Problem Analysis and Agent Design:

Problem Analysis

Educational institutions face numerous challenges in managing student inquiries efficiently. Every semester, thousands of students contact administrative departments to obtain information regarding admissions, registrations, timetables, examination schedules, attendance records, fee payments, hostel accommodation, and various campus facilities. Since these services are usually handled manually, students often experience long waiting times, delayed responses, and inconsistent information. Administrative staff spend considerable time answering repetitive questions instead of focusing on more important academic responsibilities.

An AI-powered Student Support Assistant solves this problem by automating the communication process. Instead of contacting university offices, students simply interact with the chatbot, which provides immediate responses. The chatbot works continuously throughout the day and eliminates dependency on office working hours.

Agent Objective

The main objective of the chatbot is to provide accurate, fast, and reliable answers to student queries related to university services. It aims to reduce administrative workload while improving the overall student experience.

Users

The chatbot is designed for:

- Undergraduate students
- Postgraduate students
- Faculty members
- Administrative staff
- Parents seeking academic information

Environment

The chatbot operates in an online environment and can be integrated with:

- University Website
- Student Portal
- Mobile Application
- WhatsApp

- Telegram
- Facebook Messenger

Inputs

The chatbot accepts user inputs such as:

- Text messages
- Keywords
- Course names
- Registration numbers
- Semester numbers
- Department names

Outputs

The chatbot provides:

- Course registration details
- Examination schedules
- Timetables
- Attendance information
- Campus service information
- Contact details
- Guidance for university procedures

Performance Measures

The chatbot's performance is evaluated using:

- Response accuracy
- Fast response time
- User satisfaction
- Intent recognition accuracy
- Successful query resolution
- System availability

- Error handling capability

2.Dialogflow Agent Design & Implementation

Google Dialogflow is a conversational AI platform that simplifies chatbot development. It allows developers to create intelligent conversational agents using graphical interfaces without extensive programming knowledge.

Intents

Intents represent the purpose of a user's query.

Examples include:

- Welcome Intent
- Course Registration Intent
- Timetable Intent
- Attendance Intent
- Examination Schedule Intent
- Library Services Intent
- Hostel Services Intent
- Campus Facilities Intent
- Contact Information Intent
- Default Fallback Intent

Entities

Entities identify important pieces of information.

Examples:

- Course Name
- Department
- Semester
- Date
- Examination Type

- Student ID
- Faculty Name

3.Problem Formulation and Decision Flow

The chatbot interaction can be represented as an Artificial Intelligence problem-solving process.

Initial State

The conversation begins when a student opens the chatbot and types a question.

Example:

"How do I register for courses?"

Goal State

The goal is achieved when the chatbot provides the required information accurately and the student's query is resolved successfully.

Actions

The chatbot performs several actions:

- Receive user query
- Process natural language
- Identify user intent
- Extract important entities
- Search appropriate response
- Display reply
- Wait for next query

State Transition

Initial State

↓

User enters query

↓

Intent Recognition

↓

Entity Extraction

↓

Information Retrieval

↓

Generate Response

↓

Goal Achieved

Conversation Flow

Greeting

↓

Student asks question

↓

Dialogflow identifies intent

↓

System extracts important information

↓

Database or predefined response is selected

↓

Response displayed

↓

Student satisfied

↓

Conversation ends

4.Search Strategy Application

Artificial Intelligence search algorithms help conversational agents determine the best possible response.

Two common approaches are:

Breadth First Search (BFS)

BFS explores all possible conversation paths level by level.

Advantages:

- Finds shortest conversation path
- Simple implementation
- Suitable for structured dialogue systems

Limitations:

- Consumes more memory
- Less efficient for large conversation trees

Heuristic Search

Heuristic search uses intelligent rules and confidence scores to predict the best response quickly.

Advantages:

- Faster response generation
- Higher accuracy
- Reduces unnecessary search
- Better user experience

Why Heuristic Search is Preferred

Dialogflow internally uses Machine Learning confidence scores that work similarly to heuristic search.

Example:

User: "Exam timetable"

Possible intents:

- Examination Schedule (95%)
- Timetable (60%)
- Registration (20%)

The chatbot selects the highest confidence score (95%) and provides examination schedule information.

This intelligent decision-making reduces response time and improves chatbot accuracy.

Heuristic search is therefore the most suitable approach for conversational AI because it minimizes computation while maximizing response quality.

5.Testing and Evaluation:

Testing

The chatbot was tested using various student queries.

User Query	**Expected Result**	**Status**
How do I register?	Registration steps	Correct
Show timetable	Timetable displayed	Correct
Exam dates	Examination schedule	Correct
Attendance percentage	Attendance details	Correct
Hostel facilities	Hostel information	Correct
Library timings	Library schedule	Correct

Evaluation

The chatbot demonstrated:

- High response accuracy

- Fast response time
- Good intent recognition
- Effective context handling
- Friendly user interaction
- Proper fallback responses

Future Improvements

The chatbot can be enhanced by:

- Voice-based interaction
- Multilingual support
- Integration with university databases
- Personalized recommendations
- AI-based analytics
- Mobile application support
- Real-time notifications
- Integration with Learning Management Systems (LMS)

Conclusion

The AI-Based Student Support Assistant developed using Dialogflow is an effective solution for modern universities. It automates student support services by providing instant, accurate, and intelligent responses to academic and campus-related queries. Through the use of Natural Language Processing, Machine Learning, intents, entities, contexts, and heuristic reasoning, the chatbot delivers an efficient conversational experience. It significantly reduces administrative workload while improving communication, accessibility, and student satisfaction. As Artificial Intelligence technologies continue to evolve, such intelligent assistants will become an essential part of smart educational institutions.